

Exhibit D — Market Data Snapshot

Trade date / initial fixing · 27 November 2026

Desk close · case-constructed, calibrated to August 2026 markets

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Exhibit D

Case fiction. Spot, curve, and discount factors below are **locked** for the assignment (27 November 2026 snapshot). Volatility is a **desk range** — students choose and justify a flat σ within it. Terms in Exhibit A are the **proposed** issue; fair value at 100% is expected to fall short of the 1.5% margin target.

Base-case inputs

Initial Level S_0	EUR 2318.47 (closing level, 27 November 2026) — locked
Coupon / capital barrier	$60\% \times S_0 = \text{EUR } 1391.08$ — proposed (Exhibit A)
Autocall barrier	$100\% \times S_0 = \text{EUR } 2318.47$ — proposed (Exhibit A)
Implied volatility σ	Student choice: flat 16.00–18.00% (Black / GBM); justify from context below
Teaching-note reference	17.00% flat $\rightarrow$ fair value $\approx 99.6\%$ at 100% issue
Dividend / decrement q	5.00% p.a. continuous — locked ; do not add a further dividend yield
Discounting	EUR OIS zeros in the observation-date table — locked
Funding (optional)	+25 bp to OIS zero — extra credit only; DF funded column
Process	GBM under the risk-neutral measure; 50,000 paths

Risk-neutral SDE for the **published decrement index**:

$$dS_t = (r_t - q)S_t dt + \sigma S_t dW_t^Q, \quad q = 5\%, \quad \sigma \in [16\%, 18\%].$$

Discount expected payoffs with the OIS discount factors below.

Underlying identifiers

Index	EURO iSTOXX 50 Future Healthcare Tilted NR Decrement 5%
STOXX symbol / Bloomberg	ISX5HCTD / ISX5HCTD Index
ISIN	CH1123123734
Currency / type	EUR · Price return (decrement already applied)
Index sponsor	STOXX Ltd.
Desk Bloomberg (Exhibit A)	EIIXHC5E / .EIIXHC5E

The Index replicates the **net-return** parent with a **constant 5% performance deduction** accruing daily. That drag is already in the published level S_t . Modelling $q = 5\%$ and simulating S is the correct treatment; modelling $q = 0$ would omit the decrement and overstate fair value.

Spot and context

Trade-date close S_0	2,318.47
Calibration reference	Published index $\approx$ 2,341 in mid-August 2026; 52-week high 2,351.63 (13 Aug 2026)
Case path to trade date	Modest pullback from August highs into the 27 November fixing
52-week range (context)	1,947.04 (2 Sep 2025) — 2,351.63 (13 Aug 2026)

EUR rates

Par EURIBOR swap rates (annual, 30/360 teaching convention) as of the trade-date close, calibrated to mid-August 2026 market levels (2.91% 1Y / 3.07% 5Y). Overnight: €STR 2.189%.

Tenor	Par swap	Discount factor	Zero (cont.)
€STR (overnight)	2.189%	1.00000	2.19%
1Y	2.91%	0.97172	2.87%
2Y	3.00%	0.94257	2.96%
3Y	3.03%	0.91429	2.99%
4Y	3.05%	0.88668	3.01%
5Y	3.07%	0.85955	3.03%

Zeros are bootstrapped from the par swaps (annual fixed vs annual, teaching bootstrap — not an official ICE close). Linear interpolation of the continuously compounded zero is used between pillars.

Discount factors on observation dates

Time is measured from **27 November 2026** on an ACT/365.25 basis. **DF (OIS)** is the locked discount factor for expected Index-linked payoffs. **DF (funded)** applies the +25 bp overlay.

Observation	Type	T (y)	r (cc)	DF (OIS)	DF (funded)
1 Mar 2027	Coupon	0.2574	2.36%	0.99393	0.99330
1 Jun 2027	Coupon	0.5092	2.54%	0.98717	0.98592
1 Sep 2027	Coupon	0.7611	2.71%	0.97961	0.97775
1 Dec 2027	Coupon + autocall	1.0103	2.87%	0.97143	0.96898
1 Mar 2028	Coupon	1.2594	2.89%	0.96424	0.96121
1 Jun 2028	Coupon	1.5113	2.91%	0.95692	0.95331
1 Sep 2028	Coupon	1.7632	2.94%	0.94955	0.94537
1 Dec 2028	Coupon + autocall	2.0123	2.96%	0.94222	0.93749
1 Mar 2029	Coupon	2.2587	2.97%	0.93523	0.92996
1 Jun 2029	Coupon	2.5106	2.97%	0.92809	0.92229
1 Sep 2029	Coupon	2.7625	2.98%	0.92098	0.91464
1 Dec 2029	Coupon + autocall	3.0116	2.99%	0.91397	0.90711
1 Mar 2030	Coupon	3.2580	2.99%	0.90712	0.89976
1 Jun 2030	Coupon	3.5099	3.00%	0.90015	0.89229
1 Sep 2030	Coupon	3.7618	3.00%	0.89322	0.88486
1 Dec 2030	Coupon + autocall	4.0110	3.01%	0.88638	0.87754
1 Mar 2031	Coupon	4.2574	3.01%	0.87965	0.87034
1 Jun 2031	Coupon	4.5092	3.02%	0.87281	0.86302
1 Sep 2031	Coupon	4.7611	3.02%	0.86599	0.85574
1 Dec 2031	Final / maturity	5.0103	3.03%	0.85929	0.84859

Table 1: Locked discount factors for every coupon, autocall, and final-valuation date in Exhibit A.

A constant 3.03% continuous rate (the 5Y zero) is acceptable for a **first pass**; the table above is the reference for a reproducible Monte Carlo.

Volatility

No listed options exist on ISX5HCTD. The desk works in a 16–18% flat Black vol band, consistent with:

- 17% one-year realised vol on European healthcare indices in 2026;
- VSTOXX (EURO STOXX 50, 30-day implied) in the mid-teens in August 2026.

Expiry	ATM	80% strike	60% strike
3M	15.5%	18.0%	22.0%
1Y	16.5%	18.5%	21.5%
2Y	16.8%	18.5%	21.0%
3Y	17.0%	18.5%	20.5%
5Y	17.0%	18.5%	20.5%

Student task: pick one flat ATM vol (5Y column as anchor), state your source, and run ± 2 vol-point sensitivities. The smile is **optional** extra credit; do not mix ATM and 60% vols in the same base-case path set. **Instructor reference:** $\sigma = 17.00\% \rightarrow$ proposed 6% coupon terms are **too rich** for a 1.5% margin at 100% issue.

Other desk assumptions

Repo / borrow	0 bp (cash equity index; no stock-loan)
Correlation	Not required (single underlying)
Issuer funding	+25 bp vs OIS (optional overlay; DF funded column)
Credit of hypothetical issuer	Unsecured notes; no CSA. Hedge is assumed OIS-discounted
Business days	As Exhibit A (Modified Following, Index Calculation Days)

What not to pull from a live feed

Do not replace S_0 , the discount-factor table, or the decision date with a later market print. The assignment is internally consistent only at this snapshot.

Sources for calibration (not live inputs): STOXX ISX5HCTD factsheet; EURIBOR swap indications around 17 August 2026; VSTOXX and sector realised-vol context, August 2026.